

Cardiovascular Disease Risk in Adults With Hematological Malignancies

A Population-Based Cohort Study Using Linked Databases

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ABSTRACT

BACKGROUND Survival rates for hematological malignancies have substantially improved over recent decades, prompting attention to treatment-related side effects and long-term health complications, including cardiovascular disease (CVD). These complications may limit cancer treatment tolerability and contribute to increased morbidity and mortality.

OBJECTIVES We evaluated the short- and long-term incidence of CVD across hematological malignancies.

METHODS Adult patients (aged ≥ 18 years) registered in the Netherlands Cancer Registry from 1995 to 2023 diagnosed with 1 of the 12 most common hematological malignancies were matched with general population control subjects. Data were linked to national hospitalization and cause of death registries to ascertain 11 cardiovascular outcomes. Poisson regression and Fine and Gray competing risks models were used to evaluate absolute and relative CVD risk.

RESULTS Among 174,984 patients with hematological malignancies matched with 855,085 control subjects, heart failure incidence was elevated across all malignancy subtypes, with highest excess rates per 1,000 person-years in myelodysplastic syndrome (37.75; 95% CI: 35.60-39.90) and multiple myeloma (24.68; 95% CI: 23.46-25.90). Venous thromboembolism risk peaked within the first year after diagnosis and remained elevated up to 5 years across all malignancies. First-year HRs for deep venous thrombosis ranged from 3.52 (95% CI: 2.58-4.80) in chronic lymphocytic leukemia to 34.04 (95% CI: 19.83-58.44) in Hodgkin lymphoma.

CONCLUSIONS Patients and survivors of hematological malignancies face substantially elevated short- and long-term risk of CVD compared with the general population, varying markedly by malignancy subtype. Identifying patient-, disease-, and treatment-specific drivers of this risk is essential to develop targeted prevention and management strategies to improve outcomes. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

Over the past decades, survival rates for patients with cancer have markedly improved, primarily driven by advances in early detection and therapeutic strategies.¹ Consequently, increasing emphasis has been placed on preventing and managing treatment-related side effects

and long-term health complications, including cardiovascular disease (CVD).^{2,3} Cancer patients and survivors experience a heightened risk of CVD, driven by several factors, including exposure to cardiotoxic antineoplastic agents and radiotherapy as well as shared risk factors between cancer and CVD,

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**ABBREVIATIONS
AND ACRONYMS****AML** = acute myeloid leukemia**CLL** = chronic lymphocytic leukemia**CVD** = cardiovascular disease**DLBCL** = diffuse large B-cell lymphoma**ET/PV** = essential thrombocythemia/polycythemia vera**HL** = Hodgkin lymphoma**ICD** = International Classification of Diseases**MDS** = myelodysplastic syndrome**MM** = multiple myeloma**NCR** = Netherlands Cancer Registry**NHL** = non-Hodgkin lymphoma

such as smoking, obesity, and chronic inflammation.⁴⁻⁶

To address these concerns, the emerging field of cardio-oncology has prioritized optimizing cardiovascular care for cancer patients and survivors, which is essential for improving cancer treatment tolerability and reducing long-term morbidity and mortality.³ Central to this effort is the identification of patients at high risk of CVD, enabling targeted prevention and management strategies.

Large population-based cohort studies have investigated the incidence of cardiovascular outcomes across various malignancies and risk factors associated with their occurrence.⁷⁻¹⁰ These studies demonstrated a markedly elevated risk of CVD among cancer patients relative to the general population, with substantial variation across cancer types. Notably, individuals with hematological malignancies appear to exhibit particularly high CVD risk, not only compared with the general population but also relative to patients with most solid tumors.⁷⁻¹¹

However, the limited sample sizes of prior studies have often necessitated grouping heterogeneous hematological malignancies into broad categories such as leukemia or non-Hodgkin lymphoma (NHL).⁷⁻¹⁰ This poses significant challenges for effective risk stratification, because these malignancies vary substantially in their biological characteristics, treatment protocols, and survival. To overcome the limitations of previous studies, we leveraged linked national health registries with the aim to provide robust estimates of both short- and long-term risks of diverse CVD outcomes across a comprehensive range of hematological malignancies.

METHODS

STUDY DESIGN. In this linked population-based cohort study, patients diagnosed with 1 of 12 most common hematological malignancies, including acute lymphoblastic leukemia, acute myeloid leukemia (AML), chronic lymphocytic leukemia (CLL), chronic myeloid leukemia, diffuse large B-cell lymphoma (DLBCL), essential thrombocythemia or polycythemia vera (ET/PV), follicular lymphoma, Hodgkin lymphoma (HL), indolent NHL (consisting of lymphoplasmacytic lymphoma, marginal zone lymphoma, and unspecified indolent NHL), mantle cell lymphoma, myelodysplastic syndrome (MDS), and multiple myeloma (MM), were matched with control

subjects from the general population and cancer-free control subjects to determine absolute and relative risks for 11 cardiovascular outcomes. Linked data sources covered the Netherlands between January 1, 1995 and December 31, 2023. Analyses were restricted to this setting and time period.

In accordance with guidance from the Dutch Central Committee on Research Involving Human Subjects, this observational, noninterventional study using anonymized data did not require review by a medical research ethics committee. The Privacy Review Board and Scientific Committee of the Netherlands Cancer Registry (NCR) approved the study and provided fully anonymized data. Accordingly, individual informed consent was not required. This study follows the STROBE guidelines for observational studies ([Supplemental Table 1](#)).¹²

PARTICIPANTS. The source population consisted of all patients diagnosed with a hematological malignancy in the Netherlands between 1989 and 2023 who were ≥18 years old at the time of diagnosis. Patients were identified from the NCR, maintained by the Netherlands Comprehensive Cancer Organization. The NCR captures clinical data on all patients diagnosed with cancer nationwide since 1989, including cancer diagnosis, tumor staging, topography, morphology, and first-line treatment administered within 9 to 12 months after diagnosis. Record linkage with other registries was performed for patients uniquely identifiable by the combination of date of birth, sex, and 6-character postal code corresponding to a street segment or address cluster. Linkage succeeded for 90.2% of the cohort. Because registry linkage was feasible only from 1995, the analytic cohort was restricted to patients registered in the NCR from 1995 onward. For patients with multiple records for the same diagnosis, only the record with the earliest diagnosis date was retained. Finally, before 2012, hospitalization records lacked a personal identifier. Consequently, linkage to the national hospitalization database was restricted to participants who were uniquely identifiable on the index date by the combination of date of birth, sex, and 4-character postal code representing a neighborhood or district within a city. General population control subjects were identified from the Personal Records Database, maintained by Statistics Netherlands, which contains demographic information (sex, age, and migration background) of all legal residents and registered non-residents (emigrated individuals) of the Netherlands from October 1, 1994 onward.

OUTCOMES AND PROCEDURES. Information on CVD outcomes was collected from the national

hospitalization database maintained by Dutch Hospital Data. This database captures medical and administrative information on all inpatient hospital admissions in the Netherlands from 1995 onward for individuals in the Personal Records Database. Diagnoses are coded through the International Classification of Diseases (ICD), 9th Revision (ICD-9) codes through 2012 and ICD, 10th Revision (ICD-10) codes from 2013 onward. For each admission, a primary diagnosis and up to 10 secondary diagnoses are recorded by each treating medical specialty. Study outcomes were ascertained from the primary discharge diagnosis, secondary diagnoses recorded during hospitalization, and the underlying cause of death. Cardiovascular outcomes of interest, including the ICD-9 and ICD-10 codes used to define these outcomes, are provided in [Supplemental Table 2](#).

Baseline comorbidities were defined as present if recorded during any hospitalization before the index date or inferred from relevant medication use. An updated Charlson Comorbidity Index Score was derived from hospitalization diagnoses before the index date.¹³ Causes of death, coded using ICD-10, were obtained from the National Cause of Death Registry maintained by Statistics Netherlands. Baseline medication use was obtained from the Medication Registry, which provides annual records on prescriptions reimbursed under the statutory basic health insurance for all individuals in the Personal Records Database from 2006. Medications are classified using the World Health Organization Anatomical Therapeutic Chemical system at the fourth level of classification. Baseline cardiovascular medication use was defined as having ≥ 1 relevant Anatomical Therapeutic Chemical-4 code recorded in the 2 years preceding the index year. An overview of all included variables, along with their definitions, is provided in [Supplemental Table 3](#).

STATISTICAL ANALYSIS. For all analyses, follow-up extended from the date of diagnosis (index date) until the earliest occurrence of the outcome of interest, death, emigration, loss of unique linkage status to hospitalization records, or end of study period (December 31, 2023). Categorical variables are reported as counts and percentages. Continuous variables are summarized as mean $\pm$ SD for normally distributed variables and median (IQR) otherwise. Matching was performed in a 1:5 ratio without replacement, exact on sex and 2-character postal code (corresponding to region or city area) at the index date, and nearest neighbor on year of birth.

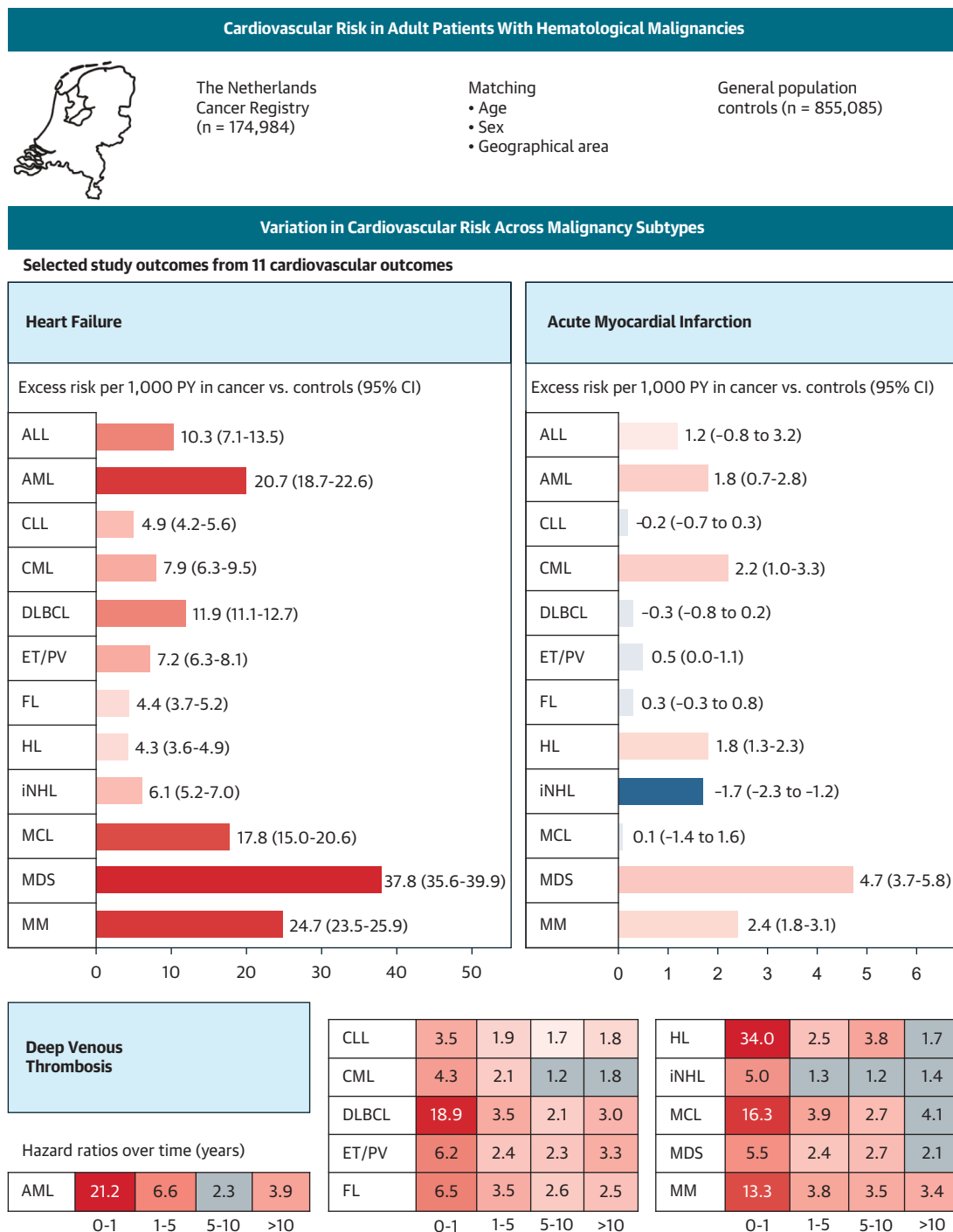
Covariate balance was assessed using standardized mean differences.

Absolute incidence rates and excess morbidity were estimated for each outcome using Poisson regression, with the number of events as the dependent variable and person-time of follow-up included as an offset in the model. For both cancer patients and control subjects, absolute incidence rates are reported as incidence rates per 1,000 person-years, and excess rates were calculated as the absolute difference in risk between these groups. Fine and Gray competing risks models, adjusting for the matching variables age and sex, were fitted separately for 4 separate time intervals to estimate HRs: <1 year, 1 to 5 years, 5 to 10 years, and >10 years after the index date. Each time interval included only individuals who were alive and event-free at the start of the interval. Death from causes other than the outcome of interest was treated as a competing event in all models.

Three sensitivity analyses were conducted. First, cancer patients with a prior cancer diagnosis recorded in the NCR were excluded. During control selection, the entire NCR database, including both hematological and solid malignancies, was used to exclude individuals with any prior cancer diagnosis from the pool of eligible control subjects. Second, patients and matched control subjects with pre-existing CVD, defined as any study outcome occurring before the index date, were excluded to differentiate between de novo disease and worsening of known baseline conditions. Then, we restricted the analysis to younger (age <70 years) cancer patients diagnosed from 2008 onward who received cancer treatment to reduce bias from untreated elderly and/or frail cancer patients. In the third sensitivity analysis, HRs were computed for 3 time intervals, <1 year, 1 to 5 years, and >5 years post-index date, because of reduced sample size and follow-up. For this analysis, relative risks were estimated using an extended Fine and Gray competing risk model adjusting for age, sex, diabetes, hypertension, dyslipidemia, and all cardiovascular comorbidities. Point estimates are presented with 95% CIs, and statistical significance was defined as a 2-sided $P < 0.05$. Data linkage and all analyses were performed in R version 4.4.3 (R Foundation for Statistical Computing).

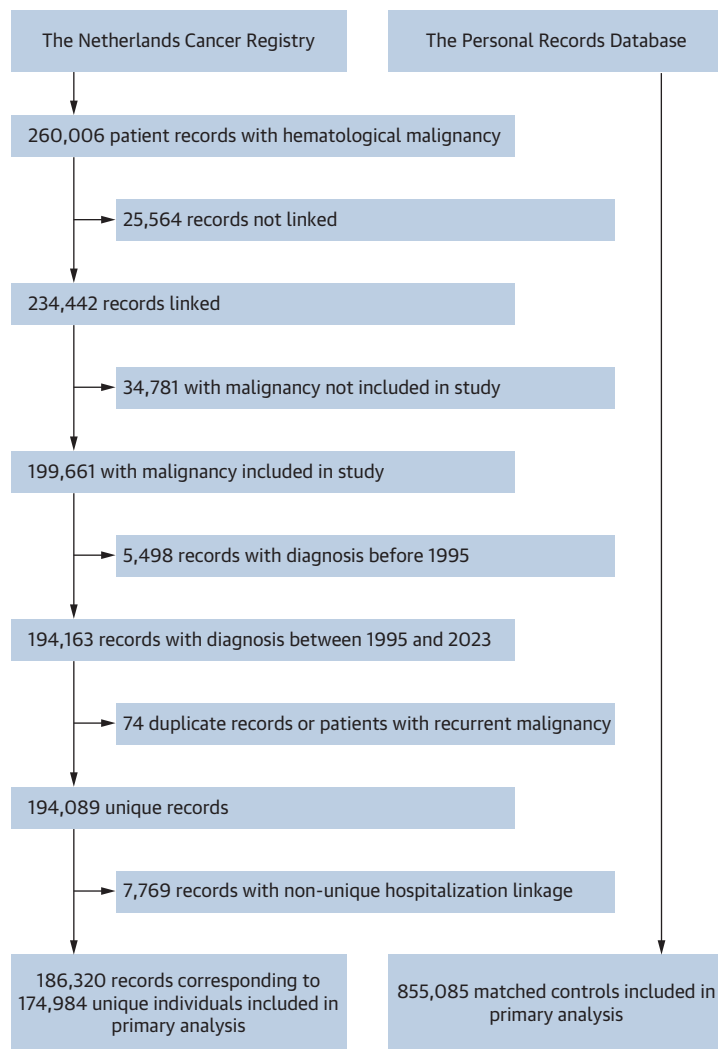
RESULTS

A total of 174,984 individuals diagnosed with a hematological malignancy were matched with 855,085 control subjects ([Central Illustration](#), [Figure 1](#)).

CENTRAL ILLUSTRATION Cardiovascular Risk in Patients With Hematological Malignancies

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Cumulative excess burdens per 1,000 person-years (PY) of heart failure and acute myocardial infarction in cancer patients vs matched control subjects are shown, with positive excess burden shown in red and negative excess burden in blue. HRs for deep venous thrombosis are shown for 0 to 1, 1 to 5, 5 to 10, and >10 years after cancer diagnosis. Box colors indicate risk magnitude and direction: red for significantly higher risk, and gray for nonsignificant differences in cancer patients vs control subjects. Abbreviations as in [Figure 2](#).

FIGURE 1 Flow Diagram of Study Cohort Selection

Baseline characteristics of all included cancer patients stratified by hematological malignancy are presented in [Table 1](#). The 3 most frequent cancer types were DLBCL (18.8%, $n = 32,877$), MM (16.9%, $n = 29,567$), and CLL (13.4%, $n = 23,412$). Overall, 12.7% of patients ($n = 22,293$) had a history of cancer, of whom 13.2% ($n = 2,939$) had received prior chemotherapy. Baseline characteristics of the total cohort and matched control subjects, of malignancy-specific cohorts with matched control subjects, and of cohorts used in each sensitivity analysis are presented in [Supplemental Tables 4, 5 to 16, and 17 to 19](#), respectively. Overall, covariates were well balanced between cancer patients and control subjects, with standardized mean differences ≤ 0.1 for all

variables except the Charlson Comorbidity Index ([Supplemental Table 4](#)). The median follow-up was 3.6 years (IQR: 1.0-8.2 years) in cancer patients and 7.1 years (IQR: 3.1-12.5 years) in control subjects.

Heart failure incidence was elevated across all malignancies compared with control subjects, with excess rates ranging from 4.26 (95% CI: 3.58-4.94) in HL to 37.75 (95% CI: 35.60-39.90) in MDS ([Figure 2](#)). Excess heart failure incidence was also notably elevated in MM (24.68; 95% CI: 23.46-25.90) and AML (20.67; 95% CI: 18.72-22.62). Deep venous thrombosis and pulmonary embolism excess rates were highest in acute lymphoblastic leukemia (12.59 [95% CI: 9.68-15.50] and 4.92 [95% CI: 2.88-6.96], respectively), AML (8.88 [95% CI: 7.81-9.95] and 5.99 [95% CI: 5.05-

TABLE 1 Baseline Characteristics of Hematological Cancer Patients

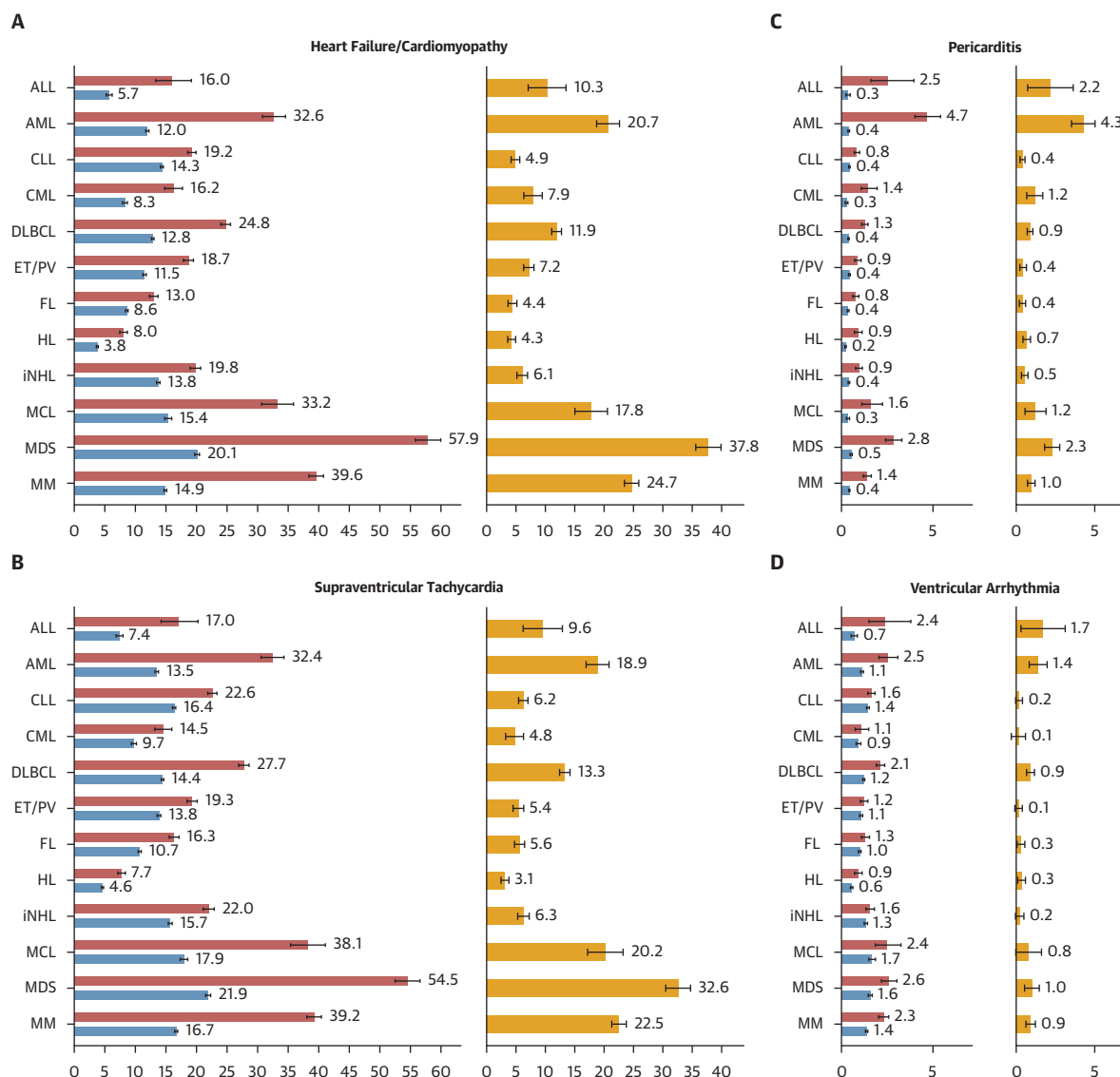
	Malignancy				
	Acute Lymphoblastic Leukemia (n = 1,877)	Acute Myeloid Leukemia (n = 16,153)	Chronic Lymphoblastic Leukemia (n = 23,412)	Chronic Myeloid Leukemia (n = 4,487)	Diffuse Large B-Cell Lymphoma (n = 32,877)
Age, y	54 (34-68)	69 (57-77)	70 (62-77)	61 (47-73)	69 (58-77)
Female	900 (47.9)	7,192 (44.5)	8,941 (38.2)	1,983 (44.2)	14,616 (44.5)
Migration background					
None (Dutch)	1,515 (80.7)	13,773 (85.3)	20,826 (89.0)	3,714 (82.8)	27,664 (84.1)
Europe	136 (7.2)	1,060 (6.6)	1,519 (6.5)	307 (6.8)	2,182 (6.6)
Other	226 (12.0)	1,320 (8.2)	1,067 (4.6)	466 (10.4)	3,031 (9.2)
Cardiovascular risk factors ^a					
Diabetes	130 (11.5)	1,405 (13.6)	2,059 (13.4)	361 (12.8)	3,139 (14.2)
Dyslipidemia	245 (21.8)	3,194 (31.0)	4,935 (32.1)	801 (28.5)	7,303 (33.0)
Hypertension	240 (21.3)	3,312 (32.2)	4,727 (30.8)	706 (25.1)	7,159 (32.3)
Comorbidities					
Arrhythmia/conduction disorder	77 (4.1)	1,367 (8.5)	1,896 (8.1)	284 (6.3)	2,893 (8.8)
Cerebrovascular accident	35 (1.9)	504 (3.1)	580 (2.5)	117 (2.6)	1,052 (3.2)
Chronic obstructive pulmonary disease/asthma ^a	139 (12.3)	1,702 (16.5)	2,555 (16.6)	489 (17.4)	3,794 (17.1)
Coronary artery disease	120 (6.4)	1,700 (10.5)	2,370 (10.1)	423 (9.4)	3,460 (10.5)
Heart failure/cardiomyopathy	32 (1.7)	561 (3.5)	680 (2.9)	145 (3.2)	1,137 (3.5)
Peripheral artery disease	20 (1.1)	336 (2.1)	406 (1.7)	82 (1.8)	703 (2.1)
Valvular heart disease	22 (1.2)	382 (2.4)	476 (2.0)	89 (2.0)	814 (2.5)
Cardiovascular drugs ^a					
Angiotensin-converting enzyme inhibitor/angiotensin II receptor blocker	244 (21.7)	3,174 (30.8)	4,916 (32.0)	772 (27.4)	6,980 (31.5)
Antithrombotics	218 (19.4)	3,430 (33.3)	4,651 (30.3)	754 (26.8)	7,302 (32.9)
Aldosterone antagonists	21 (1.9)	416 (4.0)	615 (4.0)	88 (3.1)	896 (4.0)
Beta-blockers	200 (17.8)	2,736 (26.6)	4,010 (26.1)	661 (23.5)	5,895 (26.6)
Calcium-channel blockers	105 (9.3)	1,568 (15.2)	2,259 (14.7)	364 (12.9)	3,547 (16.0)
Digoxin	≤10 (≤0.9) ^b	230 (2.2)	326 (2.1)	52 (1.8)	474 (2.1)
Diuretics	149 (13.2)	2,078 (20.2)	2,946 (19.2)	471 (16.7)	4,437 (20.0)
Insulin	40 (3.6)	397 (3.9)	603 (3.9)	122 (4.3)	957 (4.3)
Lipid-lowering agents	240 (21.3)	3,122 (30.3)	4,847 (31.6)	774 (27.5)	7,145 (32.2)
Nitrates	45 (4.0)	708 (6.9)	976 (6.4)	181 (6.4)	1571 (7.1)
Noninsulin antidiabetics	108 (9.6)	1,206 (11.7)	1,767 (11.5)	302 (10.7)	2,631 (11.9)
Charlson Comorbidity Index					
0	1,529 (81.5)	12,524 (77.5)	19,869 (84.9)	3,689 (82.2)	25,023 (76.1)
1-2	269 (14.3)	2,681 (16.6)	2,765 (11.8)	581 (12.9)	5,639 (17.2)
3-4	38 (2.0)	507 (3.1)	446 (1.9)	96 (2.1)	1,223 (3.7)
≥5	41 (2.2)	441 (2.7)	332 (1.4)	121 (2.7)	992 (3.0)
History of cancer	201 (10.7)	2,477 (15.3)	3,104 (13.3)	566 (12.6)	4,650 (14.1)
Prior chemotherapy	56 (3.0)	581 (3.6)	337 (1.4)	109 (2.4)	688 (2.1)
Prior radiotherapy	73 (3.9)	917 (5.7)	832 (3.6)	234 (5.2)	1,434 (4.4)
Current cancer treatment					
Stem cell transplantation	586 (31.2)	3,200 (19.8)	14 (0.1)	241 (5.4)	341 (1.0)
Radiotherapy	146 (7.8)	558 (3.5)	67 (0.3)	72 (1.6)	6,280 (19.1)
Chemotherapy/immunotherapy	1,596 (85.0)	11,028 (68.3)	3,324 (14.2)	4,149 (92.5)	26,848 (81.7)
Follow-up, y	1.2 (0.4-5.0)	0.5 (0.1-1.8)	5.8 (2.5-10.2)	4.9 (1.7-10.1)	2.5 (0.5-7.6)

Values are median (IQR) or n (%). ^aData were available only for patients diagnosed from 2008 onward (acute lymphoblastic leukemia, n = 1,126; acute myeloid leukemia, n = 10,299; chronic lymphocytic leukemia, n = 15,360; chronic myeloid leukemia, n = 2,815; diffuse large B-cell lymphoma, n = 22,161; essential thrombocythemia/polycythemia vera, n = 14,012; follicular lymphoma, n = 9,624; Hodgkin lymphoma, n = 5,732; indolent non-Hodgkin lymphoma, n = 11,616; mantle cell lymphoma, n = 3,222; myelodysplastic syndrome, n = 11,936; multiple myeloma, n = 19,841). ^bFor confidentiality reasons (Statistics Netherlands guidelines). ^cCell counts of 10 or fewer are reported as "≤10."

Continued on the next page

TABLE 1 Continued

Malignancy						
Essential Thrombocythemia/ Polycythemia Vera (n = 16,778)	Follicular Lymphoma (n = 14,380)	Hodgkin Lymphoma (n = 9,288)	Indolent Non-Hodgkin Lymphoma (n = 16,855)	Mantle Cell Lymphoma (n = 4,718)	Myelodysplastic Syndrome (n = 15,919)	Multiple Myeloma (n = 29,567)
68 (58-76)	64 (54-73)	41 (28-60)	70 (61-77)	71 (62-78)	75 (68-81)	71 (62-78)
9,127 (54.4)	6,990 (48.6)	4,030 (43.4)	7,449 (44.2)	1,260 (26.7)	5,972 (37.5)	12,803 (43.3)
14,032 (83.6)	12,408 (86.3)	7,387 (79.5)	14,508 (86.1)	4,157 (88.1)	13,760 (86.4)	25,258 (85.4)
1,201 (7.2)	967 (6.7)	662 (7.1)	1,028 (6.1)	286 (6.1)	1,056 (6.6)	1,800 (6.1)
1,545 (9.2)	1,005 (7.0)	1,239 (13.3)	1,319 (7.8)	275 (5.8)	1,103 (6.9)	2,509 (8.5)
1,392 (9.9)	1,076 (11.2)	362 (6.3)	1,307 (11.3)	409 (12.7)	2,468 (20.7)	2,609 (13.1)
4,834 (34.5)	2,706 (28.1)	797 (13.9)	3,246 (27.9)	1,107 (34.4)	4,983 (41.7)	6,626 (33.4)
5,032 (35.9)	2,633 (27.4)	653 (11.4)	3,497 (30.1)	1,057 (32.8)	5,200 (43.6)	7,356 (37.1)
1,501 (8.9)	938 (6.5)	276 (3.0)	1,515 (9.0)	435 (9.2)	2,621 (16.5)	2,997 (10.1)
1,214 (7.2)	334 (2.3)	102 (1.1)	431 (2.6)	132 (2.8)	797 (5.0)	867 (2.9)
2,285 (16.3)	1,551 (16.1)	768 (13.4)	2,202 (19.0)	546 (16.9)	2,809 (23.5)	3,524 (17.8)
2,067 (12.3)	1,122 (7.8)	363 (3.9)	1,563 (9.3)	534 (11.3)	3,202 (20.1)	3,243 (11.0)
559 (3.3)	303 (2.1)	103 (1.1)	576 (3.4)	146 (3.1)	1,357 (8.5)	1,336 (4.5)
680 (4.1)	195 (1.4)	85 (0.9)	318 (1.9)	100 (2.1)	690 (4.3)	552 (1.9)
395 (2.4)	234 (1.6)	70 (0.8)	495 (2.9)	122 (2.6)	991 (6.2)	912 (3.1)
4,878 (34.8)	2,617 (27.2)	654 (11.4)	3,455 (29.7)	1,065 (33.1)	4,950 (41.5)	7,211 (36.3)
6,425 (45.9)	2,585 (26.9)	757 (13.2)	3,639 (31.3)	1,124 (34.9)	5,999 (50.3)	6,831 (34.4)
603 (4.3)	282 (2.9)	75 (1.3)	491 (4.2)	109 (3.4)	935 (7.8)	1,021 (5.1)
3,958 (28.2)	2,235 (23.2)	644 (11.2)	2,929 (25.2)	908 (28.2)	4,552 (38.1)	5,936 (29.9)
2,745 (19.6)	1,295 (13.5)	328 (5.7)	1,759 (15.1)	540 (16.8)	2,330 (19.5)	3,757 (18.9)
241 (1.7)	144 (1.5)	40 (0.7)	250 (2.2)	65 (2.0)	517 (4.3)	462 (2.3)
3,098 (22.1)	1,614 (16.8)	391 (6.8)	2,223 (19.1)	631 (19.6)	3,553 (29.8)	4,599 (23.2)
393 (2.8)	297 (3.1)	115 (2.0)	352 (3.0)	100 (3.1)	706 (5.9)	726 (3.7)
4,693 (33.5)	2,641 (27.4)	772 (13.5)	3,174 (27.3)	1,086 (33.7)	4,864 (40.8)	6,475 (32.6)
947 (6.8)	508 (5.3)	155 (2.7)	747 (6.4)	223 (6.9)	1,523 (12.8)	1,411 (7.1)
1,147 (8.2)	935 (9.7)	306 (5.3)	1,086 (9.3)	362 (11.2)	2,075 (17.4)	2,194 (11.1)
13,837 (82.5)	11,839 (82.3)	8,015 (86.3)	13,231 (78.5)	3,717 (78.8)	11,017 (69.2)	22,943 (77.6)
2,210 (13.2)	1,986 (13.8)	1,020 (11.0)	2,743 (16.3)	785 (16.6)	3,375 (21.2)	4,857 (16.4)
429 (2.6)	287 (2.0)	150 (1.6)	513 (3.0)	116 (2.5)	928 (5.8)	1,038 (3.5)
302 (1.8)	268 (1.9)	103 (1.1)	368 (2.2)	100 (2.1)	599 (3.8)	729 (2.5)
1,999 (11.9)	1,887 (13.1)	575 (6.2)	2,729 (16.2)	731 (15.5)	3,114 (19.6)	3,954 (13.4)
316 (1.9)	376 (2.6)	135 (1.5)	407 (2.4)	85 (1.8)	537 (3.4)	455 (1.5)
624 (3.7)	633 (4.4)	136 (1.5)	825 (4.9)	198 (4.2)	996 (6.3)	1,174 (4.0)
≤10 (≤0.1) ^b	24 (0.2)	26 (0.3)	19 (0.1)	761 (16.1)	513 (3.2)	5,428 (18.4)
42 (0.3)	2,520 (17.5)	3,984 (42.9)	2,009 (11.9)	267 (5.7)	70 (0.4)	3,096 (10.5)
9,581 (57.1)	6,654 (46.3)	8,552 (92.1)	5,684 (33.7)	3,530 (74.8)	1,953 (12.3)	21,003 (71.0)
5.7 (2.6-9.8)	5.5 (2.2-10.8)	6.6 (2.2-13.1)	5.0 (2.0-9.3)	2.6 (0.9-5.9)	2.1 (0.8-4.8)	2.7 (0.9-5.6)

FIGURE 2 Absolute Risk in Cancer Patients and Matched Control Subjects

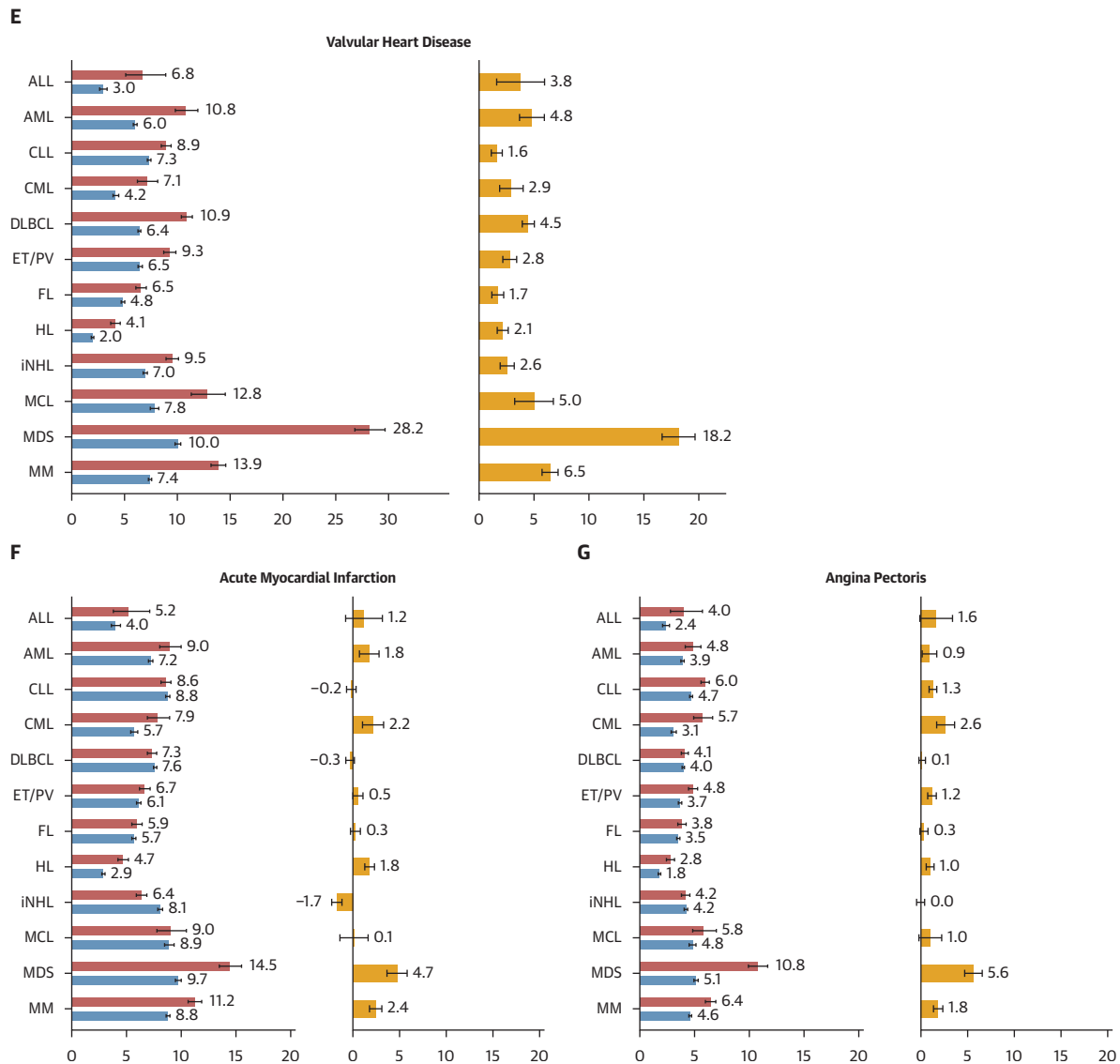
Cumulative burden per 1,000 person-years for cancer patients (red) and matched control subjects (blue), with the absolute difference illustrated in yellow. ALL = acute lymphoblastic leukemia; AML = acute myeloid leukemia; CLL = chronic lymphocytic leukemia; CML = chronic myeloid leukemia; DLBCL = diffuse large B-cell lymphoma; ET/PV = essential thrombocythemia/polycythemia vera; FL = follicular lymphoma; HL = Hodgkin lymphoma; iNHL = indolent non-Hodgkin lymphoma; MCL = mantle cell lymphoma; MDS = myelodysplastic syndrome; MM = multiple myeloma.

Continued on the next page

6.93]), and MM (4.60 [95% CI: 4.15-5.05] and 8.97 [95% CI: 8.33-9.61]). Patients diagnosed with NHL subtypes did not demonstrate significantly elevated risk of cerebrovascular accident, myocardial infarction, or angina pectoris. In patients with indolent NHL, incidence rates of cerebrovascular accident and acute myocardial infarction were lower compared with control subjects (-1.24 [95% CI: -1.85 to -0.63] and -1.72 [95% CI: -2.27 to -1.17], respectively).

The relative risk of venous thromboembolism (VTE) was highest during the first year after cancer diagnosis and remained elevated for up to 5 years across all cancer types and up to 10 years in all malignancies except chronic myeloid leukemia (Figure 3). Within the first year after diagnosis, HRs for deep venous thrombosis ranged from 3.52 (95% CI: 2.58-4.80) in CLL to 34.04 (95% CI: 19.83-58.44) in HL. For pulmonary embolism, HRs varied

FIGURE 2 Continued

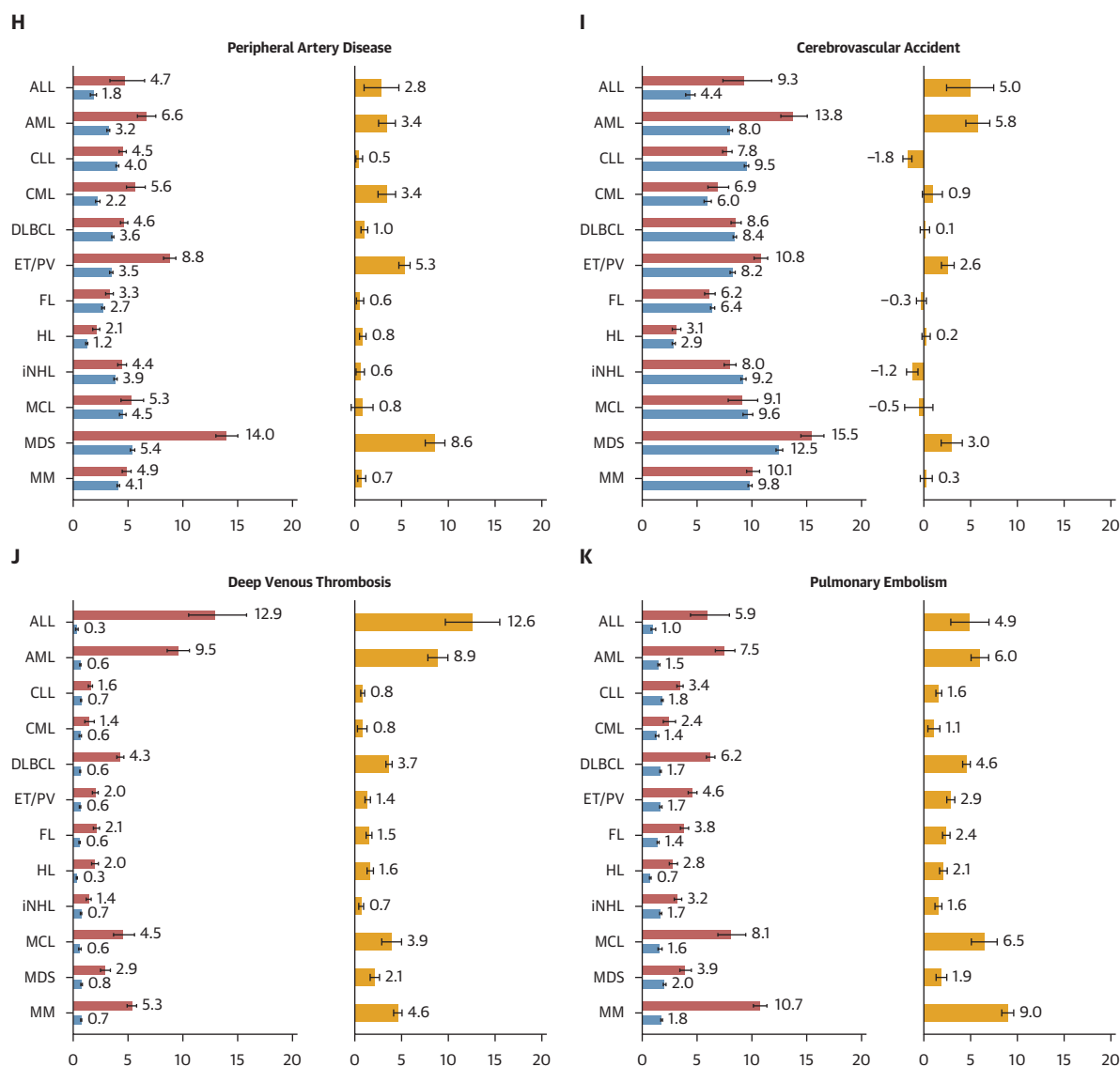


from 2.24 (95% CI: 1.70-2.95) in MDS to 23.69 (95% CI: 15.39-36.47) in HL. In patients with HL, relative risks of myocardial infarction were significantly increased until >10 years after diagnosis. Risk of cerebrovascular accident was elevated in 6 malignancies (55%) during the first year, ranging from 1.26 (95% CI: 1.11-1.42) in DLBCL to 2.39 (95% CI: 2.08-2.75) in AML. This heightened risk persisted for up to 10 years after diagnosis in patients with ET/PV and extended beyond 10 years in AML.

Sensitivity analyses in patients without a history of prior cancer yielded results that were consistent with the primary analyses (Supplemental Figures 1

and 2). Absolute excess incidence rates were overall slightly lower after excluding patients with pre-existing CVD, whereas observed HRs were higher for all time intervals and across all outcomes (Supplemental Figures 3 and 4). Finally, in treated patients aged < 70 years diagnosed from 2008 and onward, absolute CVD incidence was overall lower (Supplemental Figure 5) but relative risk estimates substantially higher (Supplemental Figure 6). Within this subgroup, heart failure risk was elevated across all 3 time periods for all malignancies. In the first year after cancer diagnosis, HRs ranged from 2.13 (95% CI: 1.55-2.92) in CLL to 17.95 (95% CI: 14.17-22.73) in

FIGURE 2 Continued



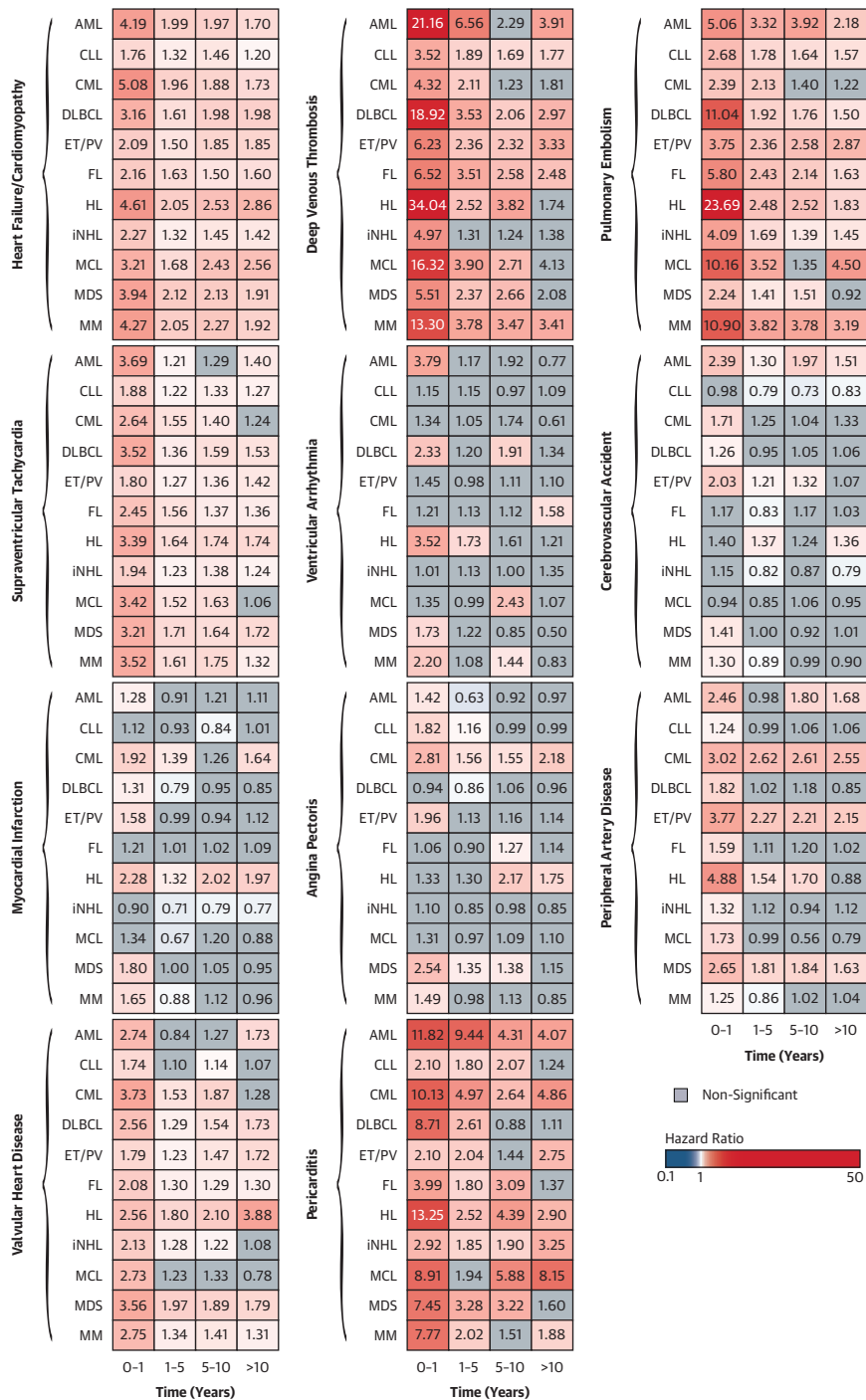
AML. Deep venous thrombosis demonstrated the highest relative risk among all outcomes, with first-year HRs ranging from 4.92 (95% CI: 2.48-9.78) in CLL to 198.41 (95% CI: 72.93-539.81) in AML. Comprehensive results of all analyses, including event rates, are presented in [Supplemental Tables 20 and 21](#).

DISCUSSION

This large population-based cohort study comprehensively characterizes short- and long-term CVD

risk in patients and survivors of hematological malignancies. Three key findings emerged. First, most CVD subtypes occurred at substantially higher rates among patients and survivors of hematological malignancies compared with matched control subjects. Second, CVD risk varied markedly by malignancy subtype and evolved over time since cancer diagnosis, underscoring the heterogeneity and temporal dynamics of CVD during cancer treatment and survivorship. Third, relative CVD risks were most pronounced in younger patients (<70 years) who received cancer treatment.

FIGURE 3 Relative Risk Over Time



HRs of cardiovascular outcomes are shown for the time periods 0 to 1, 1 to 5, 5 to 10, and >10 years after cancer diagnosis. Box colors reflect the relative magnitude and direction of risk: red for significantly higher, blue for significantly lower, and gray for nonsignificant differences in cancer patients vs control subjects. Acute lymphoblastic leukemia is not included because of a low number of events, but outcomes in this population are presented in [Supplemental Table 21](#). Abbreviations as in [Figure 2](#).

The increased cardiovascular risk among patients with hematological malignancies observed in this study aligns with findings from previous cohort studies that reported elevated CVD risk across various cancer types, including hematological malignancies.⁷⁻¹⁰ In a systematic literature review, we identified 19 cohort studies published in the past 10 years that investigated the incidence of cardiovascular outcomes in patients with hematological malignancies compared with the general population or cancer-free control subjects (Supplemental Systematic Review). However, these studies were constrained by modest sample sizes (ranging from 421 to 22,524 patients), which led to pooling of heterogeneous hematological disease subtypes, such as acute and chronic leukemias and various NHL subtypes into broad overarching categories. A recent systematic review and meta-analysis comprising 15 studies and 178,602 patients also evaluated CVD risk among patients with hematological malignancies compared with control subjects.¹⁴ However, there was substantial heterogeneity across the pooled studies, among other factors because of inconsistent covariate adjustment. Additionally, only 1 subgroup analysis was conducted to compare the incidence of stroke between patients with NHL and HL, because limited sample sizes precluded further subgroup analyses. Finally, aside from acute myocardial infarction, heart failure, and stroke, other cardiovascular outcomes were not assessed because of insufficient data. The aggregation of diverse hematological subtypes in prior studies limits the interpretability of their findings and obscures clinically relevant differences driven by disease biology, treatment exposures, and survival, all factors that influence cardiovascular risk. For instance, clonal hematopoiesis of indeterminate potential, a recognized risk factor for CVD, primarily precedes AML, MDS, and ET/PV,¹⁵⁻¹⁷ whereas anthracyclines, commonly used in DLBCL, HL, and acute leukemias but not in indolent lymphomas or MM, are strongly linked with cancer therapy-related cardiac dysfunction.¹⁸ These observations highlight the need for comprehensive, malignancy subtype-specific analyses to better delineate cardiovascular risk profiles across hematological malignancies.

We evaluated the risk of cardiovascular outcomes across 12 hematological malignancies, revealing considerable variation in CVD risk. For example, the absolute heart failure risk was elevated across all hematological malignancy subtypes, but excess rates ranged from 4.26 (95% CI: 3.58-4.94) in the relatively

young HL population to 37.75 (95% CI: 35.60-39.90) in MDS, with a similarly high excess rate of 24.68 (95% CI: 23.46-25.90) observed in patients with MM. The latter finding aligns with those of a population-based study from the United Kingdom, which included a cohort of 20 primarily solid malignancies among patients who survived ≥ 1 year postdiagnosis.⁸ Similarly, in a U.S. population-based study comparing the incidence of CVD across 9 cancer types, cumulative heart failure incidence was highest among patients with MM.¹¹ In contrast to MM, and to the best of our knowledge, data on heart failure incidence in patients with MDS have not yet been reported. A population-based cohort study reported substantially higher cardiovascular mortality in MDS patients compared with age-matched control subjects (standardized mortality ratio: 2.21; 95% CI: 2.09-2.32).¹⁹ Another retrospective study among U.S. Medicare beneficiaries similarly found a 73.2% incidence of cardiac-related events during a 3-year follow-up in patients with MDS, compared with 54.5% in the Medicare general population (age-adjusted OR: 2.10; $P < 0.01$).²⁰ Similar to absolute risks, relative risks of heart failure were also elevated across all hematological malignancy subtypes in the current study, typically peaking within the first year after cancer diagnosis, declining thereafter, and remaining elevated beyond 10 years.

Absolute incidence rates of myocardial infarction and cerebrovascular accident were not significantly elevated in patients with NHL subtypes, with only a transient increase in the relative risk during the first year after cancer diagnosis in patients with DLBCL. Relative to control subjects, patients with indolent NHL showed significantly lower absolute incidence rates for these outcomes and a lower relative risk after 1 year. Findings from other population-based studies have been inconsistent regarding the risk of these outcomes in this patient group. One study found elevated risks for stroke and ischemic heart disease over 5 years, albeit not statistically significant (adjusted HR: 1.16 [95% CI: 0.99-1.36] and 1.03 [95% CI: 0.87-1.23], respectively), whereas another study observed an increased stroke risk but not myocardial infarction over a median follow-up of 11.8 years (adjusted HR: 1.58 [95% CI: 1.45-1.72] and 1.05 [95% CI: 0.88-1.26], respectively).^{7,9} Findings from another population-based cohort study indicated higher risks of coronary artery disease and stroke among NHL patients surviving ≥ 1 year compared with cancer-free control subjects (adjusted HR: 1.30 [95% CI: 1.10-1.54] and 1.48 [95% CI: 1.24-1.76],

respectively).⁸ A systematic review and meta-analysis pooling results of 6 studies reported an elevated risk of stroke among NHL patients (HR: 1.31; 95% CI: 1.04-1.64).¹⁴

Another key finding of the current study was the temporal variation in cardiovascular risk. Relative risk of VTE was particularly elevated during the first year after diagnosis, with HRs for deep venous thrombosis ranging from 3.52 (95% CI: 2.58-4.80) in CLL to 34.04 (95% CI: 19.83-58.44) in HL and for pulmonary embolism from 2.24 (95% CI: 1.70-2.95) in MDS to 23.69 (95% CI: 15.39-36.47) in HL. Similar outcomes have been reported in a study using Danish data, with 1-year HRs for VTE of 60.7 (HL) (95% CI: 21.7-169.2), 13.4 (MM) (95% CI: 9.8-18.3), 14.8 (NHL) (95% CI: 12.0-18.1), and 6.5 (leukemia) (95% CI: 5.2-8.2).²¹ VTE risk management in hematological malignancies varies considerably across subtypes and treatment phases. In MM, thromboprophylaxis is routinely administered during first-line therapy but remains largely confined to aspirin or low-molecular-weight heparin.²² Use of direct oral anticoagulants remains limited, and prophylaxis is frequently discontinued in subsequent lines of therapy despite potentially persistent thrombotic risk, as suggested by findings from the present study. In contrast, the lower VTE burden observed in ET/PV may reflect early initiation and sustained use of antiplatelet therapy alongside systematic cardiovascular risk management.²³ However, the present analyses did not incorporate established thrombotic risk stratification tools for ET/PV, which may affect this interpretation.²⁴ These observations raise the question of whether similar longitudinal prophylactic strategies merit investigation in other hematological malignancies.

In the sensitivity analysis restricted to younger (aged <70 years) treated cancer patients, absolute CVD incidence rates were lower compared with the overall cohort, reflecting a healthier population. However, relative risk estimates were substantially more pronounced in this subgroup, likely due to reduced confounding from age-related comorbidities, thereby more clearly demonstrating cardiovascular risk mediated by cancer and its treatment. Moreover, younger patients survive longer and remain at risk for CVD over extended durations, potentially amplifying relative risks compared with the general cohort, which included older frail patients with poor prognosis and early competing mortality. Furthermore, health policy may prioritize CVD risk management and preventive strategies in younger patients with better cancer

prognosis, altering both event detection and true CVD risk.

STUDY LIMITATIONS. This nationwide study quantifies CVD risk across the full spectrum of hematological malignancies. The large sample size enabled both cancer subtype- and outcome-specific estimates without pooling biologically heterogeneous cancers or relying on composite cardiovascular endpoints, yielding granular and clinically relevant results. The analysis captures risk both during cancer treatment and into survivorship. Moreover, the use of national registries enhances representativeness, and the inclusion of patients with a history of cancer or CVD reflects real-world practice. However, this study also has some limitations. First, data on some cardiovascular risk factors such as smoking status, obesity, physical activity, and family history were unavailable. Despite adjustment for age and sex in all analyses and for cardiovascular risk factors and comorbidities in sensitivity analyses, residual confounding due to unmeasured or incompletely captured risk factors and comorbidities cannot be excluded. Second, detailed treatment data were not included to maximize follow-up duration, because treatment data granularity evolved significantly over time. Although cancer treatment is a major contributor to cardiovascular risk in patients with hematological malignancies, treatment allocation is strongly influenced by patient- and disease-specific characteristics. Including detailed treatment data would have introduced substantial variability, complicating the interpretation of cardiovascular outcomes across different treatment eras. Rigorous assessment of treatment-related cardiovascular risk requires detailed evaluation within each individual malignancy, with careful adjustment for confounding factors. However, the sensitivity analyses restricted to younger patients who received cancer treatment in the last 15 years provides CVD risk estimates more representative of patients who are potentially eligible for intensive therapy in contemporary practice.

CLINICAL IMPLICATIONS FOR CARDIOVASCULAR SURVEILLANCE AND RISK STRATIFICATION. Our findings extend existing evidence by demonstrating substantial heterogeneity in CVD risk across the most common hematological malignancies. This might have several implications for future research and clinical practice. The substantial variation in long-term CVD risk across malignancy types underscores the need for CVD risk prediction tools and targeted

cardiovascular screening strategies for cancer survivors, accounting for malignancy subtype. Currently, recommendations from the European Society of Cardiology advise annual cardiovascular risk assessment and optimization of risk factors in cancer survivors, often coordinated through primary care.³ For patients at moderate risk of cancer therapy-related cardiac dysfunction, follow-up echocardiography is advised every 5 years, whereas those at high or very high risk should be monitored at 1, 3, and 5 years post-treatment. However, baseline risk stratification is based on expert opinion rather than externally validated prediction models, and classification is determined predominantly by patient- and treatment-related factors, without consideration of malignancy subtype.²⁵ A recent systematic review identified significant gaps in cancer therapy-related cardiac dysfunction risk prediction among patients with cancer, noting that hematological malignancies and most solid tumors remain under-represented in both the development and validation of cancer therapy-related cardiac dysfunction risk prediction models.²⁵ Existing studies have focused predominantly on breast cancer populations. Furthermore, traditional cardiovascular risk scores developed for the general population, including Systematic COronary Risk Evaluation 2 (SCORE2) and QRISK3, appear to underestimate CVD risk in survivors of hematological malignancies.²⁶

In the context of atherosclerotic CVD, commonly used tools such as the American College of Cardiology/American Heart Association 2013 pooled cohort equations, which informed the development of the more recent Predicting Risk of Cardiovascular Disease Events equations, have been shown to overestimate the risk of atherosclerotic CVD in cancer survivors.²⁷⁻²⁹ The marked heterogeneity in atherosclerotic CVD risk observed in our study suggests that either malignancy-specific recalibration of existing scores or the development of cancer-specific prediction models may be required to achieve reliable risk estimates. Current cardio-oncology guidelines by the European Society of Cardiology recommend screening for coronary artery disease in high-risk patients who have received radiotherapy.³ Although the effect of radiotherapy on coronary artery disease risk was not directly assessed in the present study, we observed only a modest increased risk across most malignancies. Only in patients with HL was there a persistently elevated risk of myocardial infarction on the long term.

With growing pressures on health care systems worldwide, prediction models that account for highly differing cancer-related prognoses are needed to accurately identify patients at highest risk of CVD and focus resources where they will have the greatest impact. Large datasets capturing detailed treatment, patient, and disease characteristics could aid the development of these robust CVD risk prediction models, which subsequently need to be validated in a prospective setting and evaluated through implementation research.

CONCLUSIONS

Patients and survivors of hematological malignancies face a markedly elevated risk of CVD compared with the general population, with substantial variation across malignancies. These insights underscore the need for future research to elucidate patient-, treatment-, and disease-specific determinants of CVD risk and to inform tailored preventive and management strategies within this diverse patient population.

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KEY WORDS cardiovascular diseases, heart failure, hematological neoplasms, myocardial infarction, stroke, venous thromboembolism

APPENDIX For supplemental tables and figures and an expanded systematic review, please see the online version of this paper.